Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/25/2019
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Project Title:	Exosomes and Secretory Factors as Mediators of Non-targeted Effects of HZE Particles		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30322-4218	**Congressional District:**	5
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/6/2019 per NSSC information (Ed., 4/12/18)		
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Task Description:	This is a proposal devoted to understanding the mechanism of radiation-induced non-targeted effects, and in particular, how they lead to genomic instability in bystander cells (those not directly traversed by a radiation track). Once a bystander cell has become unstable, it is susceptible to mutations induced by endogenous and exogenous stressors, which in turn creates a permissive environment for acquisition of oncogenic mutations. Nontargeted effects (NTE) are a crucial mechanism contributing to solid tumor carcinogenesis in the space environment, yet in many respects they remain poorly understood. We hypothesize that NTE signaling leads to reliance on low-fidelity repair pathways in bystander cells; that the NTE signal is transmitted by exosomes, soluble secretory proteins, or both; and that the NTE signal affects repair through changes in the proteome and transcriptome. To test these hypotheses, we propose three specific aims. Aim 1 is to determine effects of radiation quality, dose, and dose rate on genomic instability in bystander cells, as measured by use of low-fidelity repair pathways. Experiments will focus on range of energies for intermediate Z particles. Experiments will use human bronchial epithelial cells, believed to be the target for lung carcinogenesis and bladder cancer cells, used as a positive control because they are known to generate strong non-targeted effect signals. Aim 2 is to determine the non-targeted effect signaling mechanism, with a focus on exosomes and secreted soluble proteins as candidates. Aim 3 is to investigate the exact mechanism by which NTE signaling produces genome instability in bystander cells. In summary, work will improve our quantitative understanding of radiation-induced genome instability, a major mechanism contributing to solid cancer risk, and specifically determine how genomic instability arises as a nontargeted effect in cells not directly traversed by a radiation track.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a project to characterize non-targeted effects of space radiation, that is, effects that are experienced by cells that are not directly traversed by a radiation track. In particular, the project seeks to characterize an effect of space radiation on the DNA repair machinery. When cells that have been exposed to space radiation are allowed to recover, then challenged by introduction of new DNA damage, they have an increased likelihood of repairing that damage in a way that creates chromosomal rearrangements. The effect is seen not only in cells that have been directly irradiated, but is also communicated to their non-irradiated neighbors. This non-targeted effect amplifies the risk of the original exposure. It is important to know how the effects are transmitted and whether transmission can be blocked, perhaps by pharmacological interventions. Prior studies of this phenomenon used genes that had been artificially introduced into the target cells, which created a fluorescent signal when they underwent rearrangement. In a major step forward, more advanced technology has been used to demonstrate that the same effect occurs with endogenous cancer genes in human bronchial epithelial cells, which are thought to be the relevant target cells in lung carcinogenesis. Experiments have now been performed with several different types of space radiation, in order to represent the complex environment in a shielded space habitat. One of the novel aspects of the project is to investigate whether exosomes transmit this non-targeted effect. Exosomes are nanometer-scale, membrane-bounded vesicles capable of carrying information between cells in the form of proteins, RNA, and DNA. It is important to note that radiation non-targeted effects, although harmful in the context of space radiation exposure, can be beneficial in the context of cancer therapy, where they are believed to stimulate the ability of immune cells to attack a patient's tumor. Thus, in addition to bringing about a better understanding of how non-targeted effects increase the risks of space radiation, the work may also provide insights into how to make radiation therapy more effective, particularly for late-stage metastatic cancers that are currently difficult to treat.
Task Progress:	High charge and energy (HZE) ions are a particularly harmful component of galactic cosmic radiation. In this study, the investigators tested a hypothesis that exposure to HZE ions degrades cellular DNA repair capacity. Because of this effect, human cells are less able to cope with subsequent damage incurred from exogenous or endogenous sources and more likely to become cancerous. Previously, the investigators reported that HZE ion exposure degrades repair capacity in a genetically modified human reporter cell line. The phenomenon occurs both in directly irradiated cells and in co-cultured, radiation-naïve bystanders, suggesting that the effect can be transmitted via cell-cell communication. One of the limitations of the prior work was the use of the artificial reporter cell line. Goals of the current project included examination of repair capacity at an authentic oncogene locus in a more lung cancer-relevant cell model. The investigators used RNA-guided nucleases (also known as CRISPR/Cas9) to measure an aberrant repair event involving the endogenous ALK oncogene locus. The nucleases create simultaneous, targeted, DNA double-strand breaks in the ALK and nearby EML4 loci. While in many instances the breaks are quickly and accurately rejoined (resulting in regeneration of the original chromosome sequence), occasional mis-joining creates an oncogenic EML4-ALK fusion allele. The assay mimics the single most common gene rearrangement in human lung cancer. Experiments were performed using HBEC-3KT F25F human bronchial epithelial cells, which are transformed and immortalized, but non-tumorigenic. Cultures were irradiated with HZE ions (titanium, silicon, or oxygen nuclei) delivered at various doses and energies. Cells were allowed to recover, then challenged by induction of breaks at the EML 4 and ALK loci. Results showed that a history of energetic heavy ion exposure increased the likelihood that subsequent DNA breaks will be mis-joined by up to 10-fold. There was a dose-dependent effect with titanium ions and a dose-independent effect with silicon ions. The increase in mis-joining was similar using ions of different energies. There was no evidence of elevated mis-joining with oxygen ions or with γ -rays, which were used as a reference radiation. This indicates that the effect may be limited to heavier ions (atomic number greater than 14). Overall, the results confirm the starting hypothesis that HZE ion exposure degrades repair capacity in human cells. One hypothesis to explain the degradation of repair capacity (and the increase in mis-joining) is that there is a switch from a relatively accurate pathway of DNA double-strand break repair (canonical non-homologous end joining repair, or c-NHEJ) to a mechanistically distinct and less accurate pathway (alternative nonhomologous end joining, or a-NHEJ). To investigate this, the investigators performed DNA sequence analysis of mis-repair products. Indeed, EML4-ALK junctions from HZE ion-irradiated cultures were far more likely to contain deletions, sometimes flanked by short microhomologies, than equivalent samples from mock-irradiated samples. Deletions with flanking microhomology are characteristic of a-NHEJ repair. The results thus confirm the hypothesis that there is a switch from a more accurate to a less accurate pathway of DNA repair.

	One of the limitations of the assay was variability between replicate cultures. Because of this variability, the investigators were unable to detect effect sizes of less than 5- to 10-fold. Possibly because of this limitation on sensitivity, they were unable to detect significant increases in mutagenic end joining in co-cultured, radiation-naïve bystander cells. Bystander experiments were therefore not pursued further. Exosomes are membrane-bounded vesicles that are released by cells and tissues in response to radiation and other stressors. They have a unique ability to transport DNA, RNA, proteins, and metabolites between cells. Some prior work has implicated exosomes in transmission of radiation effects between cells. To determine whether HZE ion irradiation stimulates exosome production, HBEC-3KT F25F bronchial epithelial cells were exposed to the same ions as above (titanium, silicon, or oxygen), or to γ -rays, and exosomes were prepared from the culture medium. HZE ion irradiation increased exosome secretion 2.4- to 3.4-fold, based on levels of an exosome protein marker, TSG101. Irradiation with γ -rays had no effect. Single particle analysis and electron microscopy confirmed that the exosomes were in the expected 50-150 nm size range. Levels of two damage-associated molecular patterns (DAMPs), HSP70 and calreticulin, were also seen to increase in exosomes prepared from cultures exposed to charged particle radiation. To evaluate whether there were qualitative differences between exosomes from irradiated and non-irradiated cells, the investigators analyzed microRNA (miRNA) content using NanoString technology, which allowed for parallel quantification of more than 800 well-characterized miRNAs. They compared results for exosomes from HZE ion-irradiated cultures (n=4) with mock- or γ -ray irradiated cultures (n=6). Although there were no significant differences at the individual miRNA level, further analysis using the miRNA Enrichment Analysis and Annotation (miEAA) tool identified highly significant differences at the pathway level. Specifically, cancer, DNA damage response, and senescence pathways were highly over-represented. The miRNAs in these pathways include miR-25, miR-21, mir-23a, miR-20a, miR-106a, and miR-17, which have been implicated variously in lung cancer tumorigenesis, tumor burden, and metastatic progression. These studies represent the first characterization of exosomes from HZE particle-irradiated cells.
Bibliography Type:	Description: (Last Updated: 08/06/2025)
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